Jawhar Ben Soussia

Data Scientist | ML & MLOps Engineer | AI Agent Developer

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WORK EXPERIENCE

Software Engineering Intern | *TANIT Company* 01/2025 – 06/2025 • Monastir, Tunisia

- Delivered full-stack features across 3 production modules (React / Node.js / FastAPI) in a 5-person Agile team, reducing deployment cycle time by 20%.
- Worked across 15+ technical competencies: REST APIs, CI/CD pipelines, SQL optimization, Scrum — resolving 30+ tracked GitLab issues.

Technical Lead | *Ste Smart Vision* 06/2024 – 09/2024 • Tunisia

- Led a 2-engineer team on network infrastructure and software systems; standardized deployment workflows, cutting incident response time by ~30%.

EDUCATION

M.Sc. Data Science & Strategic Operations Management | *ISIMM, Monastir* 09/2025 – Present

B.Sc. Software Engineering & Fundamental Computer Science | *ISIMM, Monastir* 09/2022 – 06/2025

KEY PROJECTS

StudyOS — Multi-Agent AI Study Assistant | *Python · LangGraph · CrewAI · AutoGen · RAG · MLflow* 09/2026 – Present

- Architected an end-to-end AI system ingesting 10–30 course PDFs into a ChromaDB vector store (Hugging Face bge-small embeddings) with cited RAG responses ≥ 70% relevance on 20 test questions.
- Built a stateful LangGraph workflow reducing hallucinations; CrewAI team generates study plans and 10-QCM quizzes from student notes via a RAG tool.
- Fine-tuned an open-source LLM (TinyLlama / Mistral-7B) using LoRA/PEFT on 200+ course Q&A pairs; trained a PyTorch classifier from scratch (EmbeddingBag + MLP).
- Deployed via FastAPI + Streamlit UI; experiment tracking with MLflow; CI/CD with GitHub Actions; reproducible Docker multi-stage compose stack.

Real-Time Cyber Threat Monitoring System | *AI Agent · FastAPI · Azure* 01/2026 – 03/2026

- Agentic detection system: Chrome extension + FastAPI backend serving XGBoost ML model via REST, deployed on Azure App Service with Docker + CI/CD pipeline.
- ML heuristic fusion (URL scoring + anomaly indicators): ~15% accuracy gain, 10% fewer false positives across 200+ test URLs, validated across 3 browsers.

Water & Agriculture Resource Optimization Pipeline | *Kafka · Airflow · Azure · Docker* 12/2025 – 03/2026

- End-to-end ETL/ELT pipeline ingesting 5+ satellite and weather data sources; Kafka streaming reduced processing latency by ~40% vs. batch-only.
- Deployed on Azure with Docker; orchestrated with Apache Airflow, PostgreSQL, dbt, MinIO, and Pandas — delivered to a real-time analytics dashboard.

TECHNICAL SKILLS

ML & AI: LangChain, LangGraph, CrewAI, AutoGen, RAG, LLMs, LoRA/PEFT, XGBoost, TensorFlow, PyTorch, scikit-learn, NLP, Deep Learning, Agentic AI, MLOps

Data Engineering & ETL: Apache Kafka, Apache Airflow, Spark, dbt, PostgreSQL, ChromaDB, MinIO, Pandas, SQL, Data Visualization

MLOps & DevOps: MLflow, GitHub Actions, Docker, CI/CD, Git, Azure (App Service), Kubernetes (familiar)

Backend & APIs: FastAPI, Python, Node.js, Express.js, REST APIs, MongoDB

Frontend & Mobile: React, Next.js, Tailwind CSS, Streamlit, React Native

Cybersecurity: Threat Detection, CTF Design, Cryptography, Network Security, Penetration Testing

Languages: Arabic (Native), French (Fluent), English (Professional)

LEADERSHIP & COMMUNITY

Cybersecurity Instructor & CTF Organizer | CPU ISIMM Organization & IEEE Computer Society | 09/2025 – Present

- Delivered cybersecurity awareness training to 100+ students across 5+ sessions; 95% satisfaction rate. Organized a CTF (10+ challenges) at Polytech Monastir and an IEEE workshop (40+ participants).
- Co-organized ISIMM Cyberbot Robotics Competition (50+ participants, +20% YoY growth).